

## SAINT-PIERRE, Saint Pierre and Miquelon

WMO: 718050

Lat: 46.7664N

Lon: 56.1792W

Elev: 23

StdP: 14.68

Time Zone: -3.00 (SPM)

Period: 99-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 13.7       | 16.2 | 6.5                           | 7.7 | 16.1 | 8.9 | 8.7 | 17.8 | 49.5                  | 29.1 | 46.0 | 28.8 | 20.8                  | 300  | 0.810 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.6                    | 70.0            | 64.5 | 68.1 | 63.8 | 66.2 | 63.0 | 66.4                | 68.3 | 65.0 | 66.7 | 63.8 | 65.4 | 12.5                 | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.8                            | 95.1 | 67.3 | 64.4 | 90.5 | 66.0 | 63.0 | 86.1 | 64.5 | 31.1          | 68.2 | 30.0 | 66.9 | 29.1 | 65.7 | 71.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 40.4              | 35.1 | 31.1 | 10.1                       | 75.1 | 3.7                | 2.1 | 7.4                                                | 76.6 | 5.2        | 77.9 | 3.1        | 79.1 | 0.4        | 80.6 |
|                   |      |      | 9.1                        | 68.8 | 3.6                | 1.2 | 6.5                                                | 69.7 | 4.5        | 70.4 | 2.5        | 71.1 | -0.1       | 72.0 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 43.1  | 28.6  | 27.1  | 30.0  | 36.0  | 42.6  | 49.5  | 58.0  | 62.6  | 57.1  | 49.2  | 41.4  | 33.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 12.67 | 5.77  | 5.86  | 5.24  | 3.72  | 3.59  | 4.04  | 3.89  | 3.03  | 4.43  | 4.80  | 5.18  | 5.80 | (7)  |
| (8)  |                                                                                  | HDD50       | 3484  | 665   | 640   | 620   | 421   | 229   | 56    | 0     | 0     | 4     | 75    | 260   | 514  | (8)  |
| (9)  |                                                                                  | HDD65       | 8023  | 1130  | 1060  | 1085  | 871   | 694   | 464   | 219   | 85    | 237   | 491   | 708   | 979  | (9)  |
| (10) |                                                                                  | CDD50       | 949   | 0     | 0     | 0     | 0     | 1     | 41    | 248   | 390   | 218   | 49    | 2     | 0    | (10) |
| (11) |                                                                                  | CDD65       | 13    | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 10    | 2     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH74       | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 1     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH80       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 15.0  | 19.3  | 19.6  | 17.6  | 15.0  | 12.6  | 11.3  | 10.1  | 11.1  | 13.3  | 15.0  | 16.7  | 18.4 | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 51.4  | 4.1   | 3.6   | 3.7   | 3.7   | 3.9   | 3.8   | 3.7   | 4.1   | 4.7   | 5.3   | 4.7   | 4.4  | (15) |
| (16) |                                                                                  | PrecMax     | 66.1  | 7.2   | 7.9   | 9.5   | 9.6   | 7.3   | 6.3   | 7.0   | 11.2  | 10.2  | 10.2  | 7.9   | 7.5  | (16) |
| (17) |                                                                                  | PrecMin     | 39.5  | 1.0   | 0.5   | 1.7   | 1.0   | 1.6   | 1.7   | 1.4   | 1.3   | 1.6   | 1.2   | 1.1   | 1.9  | (17) |
| (18) |                                                                                  | PrecStd     | 6.3   | 1.5   | 1.9   | 1.4   | 1.8   | 1.3   | 1.4   | 1.4   | 2.0   | 2.0   | 2.4   | 1.6   | 1.5  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 44.4  | 42.8  | 42.8  | 49.2  | 57.4  | 66.0  | 71.8  | 73.2  | 70.2  | 62.4  | 55.3  | 49.3 | (19) |
| (20) |                                                                                  |             | MCWB  | 44.3  | 42.4  | 39.9  | 44.5  | 48.4  | 57.0  | 63.0  | 66.1  | 65.3  | 60.0  | 54.1  | 48.9 | (20) |
| (21) |                                                                                  | 2%          | DB    | 41.2  | 39.4  | 40.1  | 45.0  | 53.6  | 60.9  | 68.4  | 70.9  | 67.0  | 59.9  | 52.6  | 46.1 | (21) |
| (22) |                                                                                  |             | MCWB  | 40.7  | 38.7  | 37.9  | 41.9  | 46.2  | 54.6  | 62.4  | 65.6  | 63.0  | 57.6  | 51.7  | 45.5 | (22) |
| (23) |                                                                                  | 5%          | DB    | 38.7  | 37.0  | 38.4  | 43.2  | 50.5  | 58.0  | 65.9  | 69.3  | 64.9  | 58.1  | 51.1  | 44.2 | (23) |
| (24) |                                                                                  |             | MCWB  | 38.0  | 36.2  | 36.6  | 40.9  | 45.5  | 53.3  | 61.4  | 64.7  | 61.8  | 56.2  | 50.3  | 43.6 | (24) |
| (25) |                                                                                  | 10%         | DB    | 36.6  | 34.9  | 36.7  | 41.5  | 48.4  | 55.8  | 63.7  | 67.6  | 63.6  | 56.3  | 49.4  | 42.0 | (25) |
| (26) |                                                                                  |             | MCWB  | 35.7  | 34.0  | 35.0  | 39.4  | 44.8  | 52.3  | 60.7  | 64.1  | 60.8  | 54.4  | 48.5  | 41.2 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 44.1  | 42.6  | 41.2  | 45.7  | 50.5  | 59.0  | 66.4  | 68.7  | 66.8  | 60.9  | 54.5  | 48.9 | (27) |
| (28) |                                                                                  |             | MCD B | 44.2  | 42.7  | 41.6  | 47.4  | 53.6  | 62.8  | 68.9  | 70.7  | 68.9  | 61.5  | 55.0  | 49.0 | (28) |
| (29) |                                                                                  | 2%          | WB    | 40.9  | 39.0  | 38.7  | 43.2  | 48.4  | 56.4  | 64.1  | 67.2  | 64.7  | 58.6  | 52.1  | 45.7 | (29) |
| (30) |                                                                                  |             | MCD B | 41.0  | 39.3  | 39.5  | 44.3  | 51.1  | 59.0  | 66.4  | 69.2  | 65.9  | 59.5  | 52.4  | 45.9 | (30) |
| (31) |                                                                                  | 5%          | WB    | 38.1  | 36.3  | 36.9  | 41.6  | 46.8  | 54.6  | 62.7  | 66.1  | 63.2  | 56.8  | 50.5  | 43.6 | (31) |
| (32) |                                                                                  |             | MCD B | 38.4  | 36.6  | 37.8  | 42.7  | 49.3  | 57.2  | 64.3  | 67.8  | 64.4  | 57.7  | 51.0  | 44.0 | (32) |
| (33) |                                                                                  | 10%         | WB    | 35.9  | 34.0  | 35.2  | 39.8  | 45.3  | 52.9  | 61.3  | 64.9  | 61.6  | 54.8  | 48.5  | 41.3 | (33) |
| (34) |                                                                                  |             | MCD B | 36.4  | 34.6  | 36.3  | 41.0  | 47.6  | 55.3  | 62.9  | 66.5  | 62.9  | 56.0  | 49.0  | 41.8 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 8.3   | 8.3   | 8.3   | 9.0   | 10.1  | 9.8   | 8.7   | 8.6   | 9.3   | 8.4   | 7.9   | 7.7  | (35) |
| (36) |                                                                                  |             | MCDBR | 10.9  | 11.3  | 9.6   | 11.2  | 13.8  | 14.0  | 12.2  | 10.7  | 10.1  | 9.6   | 9.5   | 10.4 | (36) |
| (37) |                                                                                  |             | MCWBR | 12.0  | 11.9  | 8.4   | 8.7   | 8.8   | 8.8   | 7.8   | 7.1   | 7.8   | 8.7   | 10.1  | 11.4 | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 11.2  | 11.4  | 9.4   | 10.4  | 11.4  | 11.8  | 9.5   | 8.8   | 8.8   | 9.0   | 9.5   | 10.3 | (38) |
| (39) |                                                                                  |             | MCWBR | 12.4  | 12.2  | 9.1   | 9.0   | 8.5   | 8.4   | 7.0   | 6.6   | 7.7   | 9.0   | 10.4  | 11.5 | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.323 | 0.322 | 0.340 | 0.367 | 0.379 | 0.403 | 0.433 | 0.406 | 0.373 | 0.356 | 0.330 | 0.316 | (40) |      |
| (41) |                                                                                  | taud        | 2.367 | 2.442 | 2.430 | 2.388 | 2.406 | 2.358 | 2.319 | 2.400 | 2.474 | 2.464 | 2.461 | 2.395 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 240   | 270   | 281   | 282   | 281   | 274   | 263   | 267   | 266   | 253   | 238   | 229   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 23    | 26    | 31    | 35    | 36    | 38    | 39    | 35    | 30    | 26    | 21    | 20    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 308   | 527   | 952   | 1341  | 1627  | 1651  | 1560  | 1422  | 1171  | 731   | 392   | 252   | (44) |      |
| (45) |                                                                                  | RadStd      | 27    | 49    | 68    | 116   | 135   | 175   | 112   | 98    | 75    | 49    | 50    | 25    | (45) |      |

## Historical Trends

|                             |     | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|-----------------------------|-----|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                             |     |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                         | (e) | (f)   | (g)     | (h)    | (i)   | (j)     | (k)   | (l) | (m)         |       |       |       |
| (46) Station Only           | N/A | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (6 neighbors) | N/A | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page